Unit 2 - Linear Algebra
2.4 Transformations

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MSDS 520: Math and Stats Foundations for Data Science
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Outline

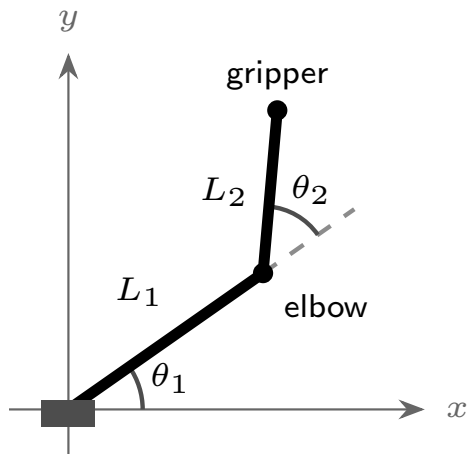
1. Matrices and Transformations
2. A Matrix as a Function
3. What the Columns Tell You
4. Types of Transformations
5. Length-Preserving Transformations
6. Composition and Order
7. Moving Away from the Origin
8. Review

Matrices and Transformations

A Problem to Motivate Us

Where is the gripper?

"A robot arm in a hospital lab has two links: an upper arm of length L_1 from its base to the elbow, and a forearm of length L_2 from the elbow to the gripper. The base turns by θ_1, and the elbow turns by θ_2 more. For **any** *angles, where is the gripper?"*



What the answer will need

- Turning about the **base** is a matrix: vector in, new vector out. This deck builds it.
- The elbow turns about the **elbow**, a point that itself moves. That is **not** a turn about the origin, so no 2×2 matrix does it.
- The last section adds one coordinate, and then the whole arm is **one product of matrices**.

Where This Shows Up in Data Science

Moving points is everywhere

- **Image augmentation.** Classifiers train on rotated, flipped, and stretched copies of every picture: one matrix applied to every pixel's coordinates.
- **Medical imaging.** Lining up two scans of the same patient (**registration**) means finding the turn and stretch that carries one onto the other.
- **Robotics and graphics.** Every robot joint and every turn of a game camera is a rotation matrix, and a chain of joints is a chain of products.
- **Changing coordinates.** Rewriting data on a better set of axes is the same product. **PCA**, this unit's capstone, does exactly that.
- **Neural networks.** Each layer applies a matrix transformation, then a simple nonlinear bend.

Unit 2-3 built the product $A\bar{x}$ as arithmetic. Today it gets a picture: what it does to **space**.

What We Will Build

The plan

1. A **matrix as a function**: a vector in, a new vector out, and why it is called **linear**.
2. **What the columns tell you**: where $\hat{\imath}$ and $\hat{\jmath}$ land, and how to build the matrix you want.
3. **Types of transformations**: scaling, shear, reflection, rotation (2D and 3D), permutation, and projection.
4. **Length-preserving** transformations: the **orthogonal** matrices.
5. **Composition and order**: two moves as one matrix, and powers.
6. **Moving away from the origin**: the moves no matrix can make, and the extra coordinate that lets matrices make them. It **solves the robot arm**.

Notebooks for this lecture: `U2-4_Transformations-1_MatrixAsMap`, `U2-4_Transformations-2_NamedTransforms`, `U2-4_Transformations-3_UnitCircleAnimation`.

A Matrix as a Function

A Matrix Is a Function

Vector in, new vector out

$$\bar{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{\text{in}} \boxed{A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}} \xrightarrow{\text{out}} A\bar{x} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

Same matrix, every input

in $\bar{x}$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 1 \end{bmatrix}$
out $A\bar{x}$	$\begin{bmatrix} 2 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 3 \\ 3 \end{bmatrix}$	$\begin{bmatrix} -1 \\ 3 \end{bmatrix}$

Just like a Unit 1 function

- $f(x) = 2x$: a **number** in, a number out. $f(3) = 6$.
- $f(\bar{x}) = A\bar{x}$: a **vector** in, a **new vector** out.
- One fixed rule, the matrix, applied to **every** input.

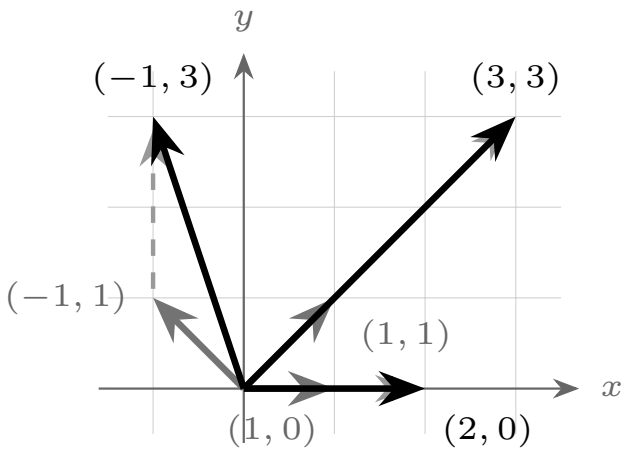
Watching a Matrix Move Vectors

Definition

Multiplying by a matrix A is a function on vectors:

$$f(\bar{x}) = A\bar{x} \qquad \text{(input: the vector } \bar{x}\text{; output: the vector } A\bar{x}\text{)}$$

A function like this is a **linear transformation**: A does the moving, sending each input vector to a new place.



Gray: inputs. Black: where $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ sends them.

What goes in, what comes out

- An $m \times n$ matrix takes vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$.
- 2×2: the plane to the plane. Today's case.
- 3×2: the plane **into** 3D space.

Notebook `U2-4_Transformations-1_MatrixAsMap`, §1, the point before and after

Why "Linear": the Two Rules and Their Consequences

The transformation respects vector arithmetic

By the distributive and scalar rules for products from Unit 2-3, for every $\bar{u}, \bar{v}$ and every scalar c,

$$A(\bar{u} + \bar{v}) = A\bar{u} + A\bar{v}, \qquad A(c\,\bar{u}) = c\,A\bar{u}.$$

Move a sum, or move the parts and add: same answer. That is what **linear** means.

Three consequences, visible in every picture today:

- **The origin stays put:** $A\bar{0} = \bar{0}$.
- **Lines stay lines**, and parallel lines stay parallel.
- **Even spacing stays even:** a square grid becomes a grid of identical parallelograms, never curved or bunched.

Ruled out: a **translation**, sliding the whole plane sideways, moves the origin, so no 2×2 matrix does it. The last section shows the extra-coordinate trick computer graphics uses to bring translations back.

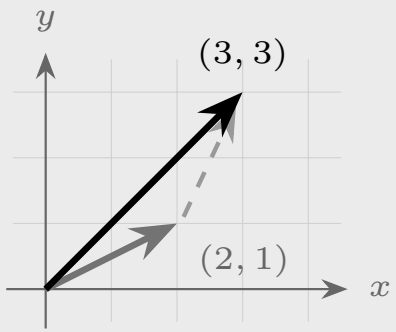
Notebook `U2-4_Transformations-3_UnitCircleAnimation`, §3, four deformed grids

One Product, Two Interpretations

The basis $\bar{b}_1 = \left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right]$, $\bar{b}_2 = \left[\begin{smallmatrix} -1 \\ 1 \end{smallmatrix}\right]$ from Unit 2-2, as the columns of B:

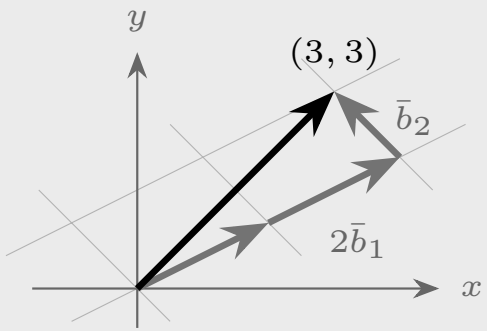
$$B \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

1. A **new vector**, same space



- Input and output are both in **standard** coordinates.
- B **moves** the point $(2, 1)$ to $(3, 3)$: two different vectors.

2. The **same vector**, new basis



- $\left[\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}\right]$ is the vector's address in the $\bar{b}$ basis: $2\bar{b}_1 + 1\bar{b}_2$.
- B **translates** that address into standard coordinates. **Nothing moved.**

PCA uses the second: it finds a better basis and rewrites every data point in it.

What the Columns Tell You

Where Do $\hat{\imath}$ and $\hat{\jmath}$ Land?

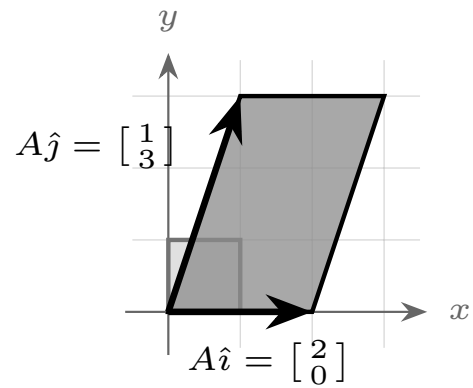
Feed the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ the two unit vectors and use the column view:

$$A\hat{\imath} = 1\begin{bmatrix} a \\ c \end{bmatrix} + 0\begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix}, \qquad A\hat{\jmath} = 0\begin{bmatrix} a \\ c \end{bmatrix} + 1\begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}.$$

The key fact of this lecture

The columns of A are the places $\hat{\imath}$ and $\hat{\jmath}$ are sent. Column 1 is the image of $\hat{\imath}$; column 2 is the image of $\hat{\jmath}$.

Why that tells you everything: every vector is a combination of $\hat{\imath}$ and $\hat{\jmath}$, and A respects combinations, so $A(x\hat{\imath} + y\hat{\jmath}) = x\,(A\hat{\imath}) + y\,(A\hat{\jmath})$. Know where the two unit arrows go and you know where **everything** goes: x of the first column plus y of the second.



$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$: unit square to parallelogram.

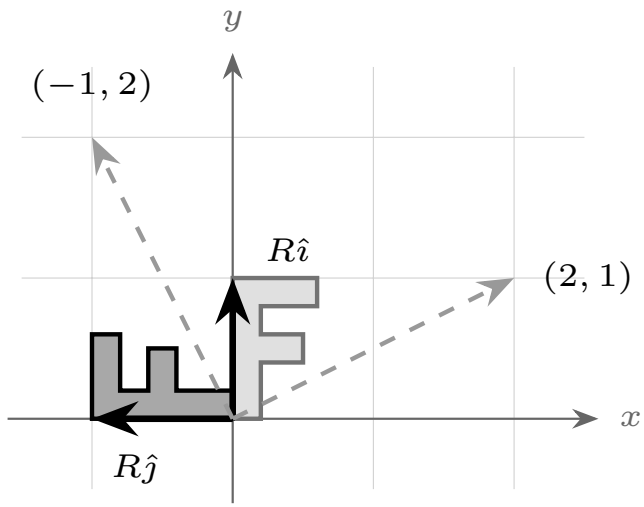
Notebook `U2-4_Transformations-1_MatrixAsMap`, §2, where the unit vectors land

Going Backwards: Build the Matrix You Want

Designing a transformation is two decisions: where should $\hat{\imath}$ go, and where should $\hat{\jmath}$ go? Write the answers as columns and you are finished.

Worked: a quarter turn counterclockwise. Turning $\hat{\imath}$ a quarter turn points it straight up; turning $\hat{\jmath}$ points it left:

$$\hat{\imath} \mapsto \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \hat{\jmath} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix} \;\Rightarrow\; R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$



The F before (light) and after R (dark); the columns as arrows.

Check on a point: $R\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, which is $(2, 1)$ turned a quarter turn, as the picture confirms.

No formula was looked up; every transformation in the next section was built this way.

Notebook `U2-4_Transformations-1_MatrixAsMap`, §4, a quarter turn built on purpose

Exercise: Read It, Then Build It

Try it

1. For each matrix, say where $\hat{\imath}$ and $\hat{\jmath}$ land and describe the transformation in words:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

2. Write the matrix that sends $\hat{\imath}$ to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\hat{\jmath}$ to $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Where does it send $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$? And $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$?
3. A matrix sends every point to a point on the x-axis. What must its second row be?

Exercise: Solution

Answers

1. **First:** $\hat{\imath}$ stays at $\begin{bmatrix}1\\0\end{bmatrix}$, $\hat{\jmath}$ goes to $\begin{bmatrix}0\\-1\end{bmatrix}$: the plane is **flipped across the x-axis**. **Second:** $\hat{\imath} \mapsto \begin{bmatrix}0\\1\end{bmatrix}$ and $\hat{\jmath} \mapsto \begin{bmatrix}1\\0\end{bmatrix}$: the two coordinates are **swapped**, a flip across the line $y = x$. **Third:** both arrows are tripled in place: everything is **scaled by** 3, directions unchanged.

2. Stack the images as columns: $A = \begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}$. It sends $\begin{bmatrix}1\\0\end{bmatrix}$ to its first column, $\begin{bmatrix}1\\1\end{bmatrix}$, and $\begin{bmatrix}2\\3\end{bmatrix}$ to $2\begin{bmatrix}1\\1\end{bmatrix} + 3\begin{bmatrix}-1\\1\end{bmatrix} = \begin{bmatrix}-1\\5\end{bmatrix}$. (This matrix turns by $45°$ and stretches by $\sqrt{2}$: both columns have length $\sqrt{2}$ and sit $45°$ from where they started.)

3. The second entry of every output is $cx + dy$, which must be 0 for all x, y. So $c = d = 0$: the **second row is all zeros**. Both columns then lie on the x-axis, which is where every combination of them must land.

Types of Transformations

Scaling and Shear

Scaling: a diagonal matrix

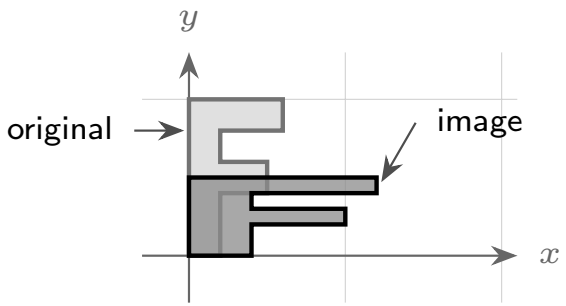
A **diagonal** matrix (nonzero only on the diagonal) stretches each axis by its own factor: $\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ doubles $\hat{\imath}$ and halves $\hat{\jmath}$.

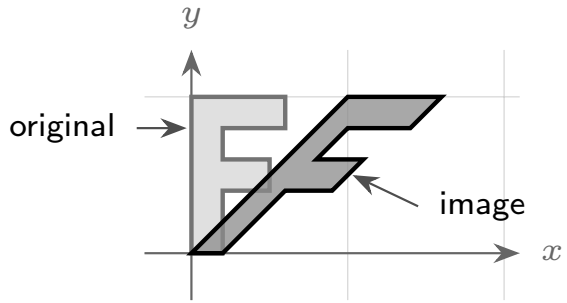
Equal factors, kI, enlarge uniformly; a negative factor flips that axis.

Shear: slide the top sideways

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: $\hat{\imath}$ stays put, $\hat{\jmath}$ tips over to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Points slide horizontally by an amount proportional to their height, like a deck of cards pushed over.





Check each matrix's columns against where the F's bottom edge and left edge went.

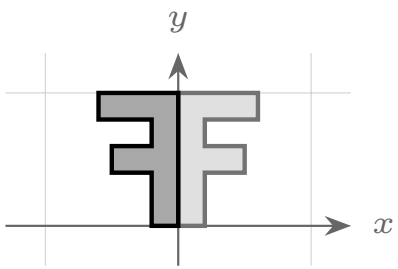
Notebook `U2-4_Transformations-2_NamedTransforms`, §1, the full panel of named transforms

Reflection

A reflection flips the plane across a **mirror line** through the origin. Build each one from where $\hat{\imath}$ and $\hat{\jmath}$ land:

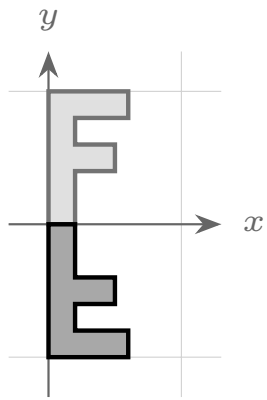
Across the y-axis

$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$: $\hat{\imath}$ reverses, $\hat{\jmath}$ stays.



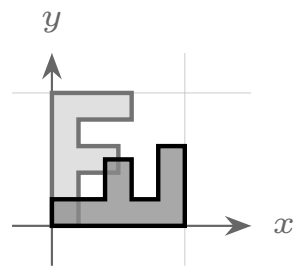
Across the x-axis

$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$: $\hat{\imath}$ stays, $\hat{\jmath}$ reverses.



Across the line $y = x$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: $\hat{\imath}$ and $\hat{\jmath}$ **trade places**.



- Flip twice and you are back: every reflection **undoes itself**.
- The last one **swaps the coordinates**. In 3D, swaps become **permutations**.

Rotation by Any Angle

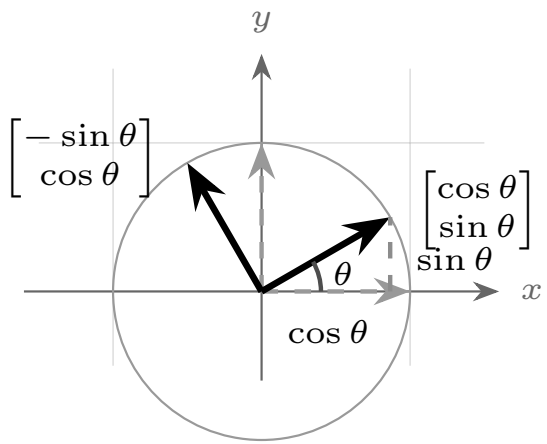
Turn $\hat{\imath}$ counterclockwise by θ. It is still a unit vector, now at angle θ, so by the definition of sine and cosine it lands at $\begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$. The vector $\hat{\jmath}$ stays a quarter turn ahead of it, landing at $\begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$.

The rotation matrix

Stack the two images as columns:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

At $\theta = 90^\circ$ this is $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, the quarter turn built by hand.



Both unit vectors turned by θ; their new coordinates are the columns.

First piece of the robot arm: turning the whole arm about its base by θ_1 is $R(\theta_1)$. The elbow needs one more idea (last section).

Rotations in 3D

In 3D a rotation turns about an **axis**. About each coordinate axis, that axis stays put and the other two turn exactly as in 2D ($c = \cos\theta$, $s = \sin\theta$):

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}, \qquad R_y = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix}, \qquad R_z = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Same column rule

- R_z's last column is $\hat{k}$: the z-axis stays put.
- Its top-left corner is $R(\theta)$ from the plane.
- $R_z(90°)$: $\hat{\imath} \mapsto \hat{\jmath}, \quad \hat{\jmath} \mapsto -\hat{\imath}$.

Where you meet them

- **Yaw**, **pitch**, **roll**: aircraft, drones, phone sensors, game cameras.
- **Order matters**: tip a book then turn it, or turn then tip: two different poses.

The sign flip in R_y is not a typo: it keeps every turn counterclockwise when you look down the axis toward the origin.

Permutations: Reordering Coordinates

A **permutation matrix** is I with its rows reordered: it reorders a vector's entries. The plane has just the swap; 3D has $3! = 6$ orderings.

A cycle is a rotation

$$P \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ c \\ a \end{bmatrix}$$

Three times brings every entry home: $P^3 = I$, a turn by $120°$ about the line through $(1, 1, 1)$.

A swap is a reflection

$$S \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ a \\ c \end{bmatrix}$$

Twice brings it back: $S^2 = I$, the mirror across the plane $x = y$.

Already used: the row swap in Gaussian elimination (Unit 2-1)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \left[\begin{array}{cc|c} 0 & 2 & 4 \\ 1 & 1 & 3 \end{array}\right] = \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 4 \end{array}\right]$$

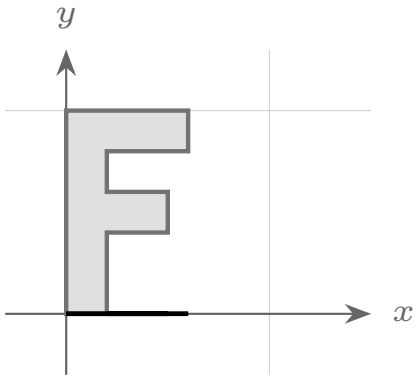
$R_1 \leftrightarrow R_2$, done to get a zero out of the pivot spot, **is** a permutation matrix on the left. Shuffling a data table's rows is the same idea.

Projection: Flattening onto a Line

Projection onto the x-axis

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

It keeps $\hat{\imath}$ and crushes $\hat{\jmath}$ to $\bar{0}$: every point drops straight down onto the x-axis.



Projection destroys information

- The whole F lands on a segment: many inputs share each output.
- So from the segment there is **no way to recover** the F, and the move cannot be undone. The identity, by contrast, moves nothing at all.
- Unit 2-5 measures that collapse with the **determinant** and asks when a transformation can be reversed with an **inverse**.

Next: the x-axis is just one line. The **outer product** builds the projection onto the line through **any** vector.

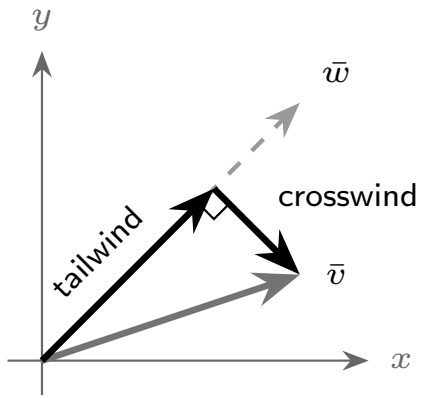
Projection onto Any Line: the Outer Product

So far we have only projected onto the x-axis. Unit 2-2 projected onto a **slanted** line.

Recall from Unit 2-2: wind $\bar{v}$, heading $\bar{w}$

$$\bar{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \hat{w} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

- **Tailwind**: $(\bar{v} \cdot \hat{w})\,\hat{w} = 2\sqrt{2}\,\hat{w} = \left[\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right]$
- **Crosswind**: $\bar{v} - \left[\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$



Regroup the same arithmetic as a matrix

The dot product is a row times a column, $\bar{v} \cdot \hat{w} = \hat{w}^\top \bar{v}$, a 1×1 number, so it can sit on either side of $\hat{w}$. Then regroup (products are associative):

$$(\bar{v} \cdot \hat{w})\,\hat{w} \;=\; \hat{w}\,(\hat{w}^\top \bar{v}) \;=\; (\hat{w}\,\hat{w}^\top)\,\bar{v} \;=\; P\bar{v}.$$

Row times column is a number (**inner** product); **column times row** is a 2×2 **matrix**, the **outer product**.

The Tailwind Matrix and the Crosswind Matrix

Tailwind: $P = \hat{w}\,\hat{w}^\top$, the projection onto the line through $\bar{w}$

Entry (i, j) of the outer product is $\hat{w}_i\,\hat{w}_j$, and here every one is $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$:

$$P = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$P\bar{v} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/2 + 1/2 \\ 3/2 + 1/2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Crosswind: $I - P$, because $\bar{v} - P\bar{v} = I\bar{v} - P\bar{v} = (I - P)\,\bar{v}$

$$I - P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix}$$

$$(I - P)\,\bar{v} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/2 - 1/2 \\ -3/2 + 1/2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Same tailwind $\left[\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}\right]$ and same crosswind $\left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$ as Unit 2-2. And $P + (I - P) = I$: every vector splits into tailwind plus crosswind.

Two Ways to Project

	Dot-product way (Unit 2-2)	**Matrix way** (today)
Tailwind	$(\bar{v} \cdot \hat{w})\,\hat{w}$	$P\bar{v}, \quad P = \hat{w}\,\hat{w}^\top$
Crosswind	$\bar{v} - (\bar{v} \cdot \hat{w})\,\hat{w}$	$(I - P)\,\bar{v}$
The work	redone for **each** vector	build P **once**, apply to all

What the matrix way buys

- **A transformation** like the others in this section: one matrix moves every point at once.
- **Projection onto the x-axis is the special case** $\hat{\imath}\,\hat{\imath}^\top = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$: the shadow on the x-axis.
- $I - P$ **is a projection too**, onto the perpendicular line through $\left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$. P forgets the crosswind; $I - P$ forgets the tailwind.
- **Read the columns**: both are $\begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$, so $\hat{\imath}$ and $\hat{\jmath}$ both land on the line through $\bar{w}$.

Exercise: Types of Transformations

Try it

1. Rotate the point $(2, 0)$ by $90°$ counterclockwise using $R(90°)$. Then rotate $(1, 1)$ by $45°$ using $R(45°)$ (take $\cos 45° = \sin 45° = \frac{1}{\sqrt{2}}$). Sketch both.
2. Apply the shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ to the four corners of the unit square, $(0, 0)$, $(1, 0)$, $(1, 1)$, $(0, 1)$, and describe the shape you get.
3. Write the matrix that reflects across the line $y = -x$, and check it on $(3, 5)$.
4. Where does the projection $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ send $(3, 5)$, $(3, -2)$, and $(3, 0)$? What does that tell you about undoing it?
5. Apply the cycle $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ to $(1, 2, 3)$, then to the result, then once more. What happens?

Exercise: Solution

Answers

1. $R(90°)\begin{bmatrix}2\\0\end{bmatrix} = \begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}\begin{bmatrix}2\\0\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix}$: from the x-axis straight up to the y-axis. $R(45°)\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1-1\\1+1\end{bmatrix} = \begin{bmatrix}0\\\sqrt{2}\end{bmatrix}$: the diagonal point turns onto the y-axis, keeping its length $\sqrt{2}$.
2. $(0,0) \mapsto (0,0)$, $(1,0) \mapsto (1,0)$, $(1,1) \mapsto (2,1)$, $(0,1) \mapsto (1,1)$. The bottom edge stays, the top edge slides one unit right: a **parallelogram** leaning right, with the same base and height as the square.
3. The mirror $y = -x$ sends $\hat{\imath}$ to $\begin{bmatrix}0\\-1\end{bmatrix}$ and $\hat{\jmath}$ to $\begin{bmatrix}-1\\0\end{bmatrix}$, so the matrix is $\begin{bmatrix}0 & -1\\-1 & 0\end{bmatrix}$. It sends $(3,5)$ to $(-5,-3)$: swap the coordinates and negate both.

Exercise: Solution (continued)

Answers

4. All three go to $(3, 0)$. Three inputs, one output, so **no function can send $(3, 0)$ back** to the right one: the projection cannot be undone. That is what "destroys information" means, and a zero determinant will detect it in Unit 2-5.
5. Applying P three times: $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \to \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} \to \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \to \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Each step moves every entry up one place, the top one wrapping to the bottom. After three steps it is **back where it started**: $P^3 = I$, three turns of $120°$ making a full turn.

Length-Preserving Transformations

Which Transformations Keep Lengths?

The question: when is $\|A\bar{x}\| = \|\bar{x}\|$ for every $\bar{x}$? (A rotation: yes. A shear: no.)

The test: $A^\top A = I$

$$\|A\bar{x}\|^2 = (A\bar{x})^\top (A\bar{x}) = \bar{x}^\top \underbrace{A^\top A}_{=I} \bar{x} = \bar{x}^\top \bar{x} = \|\bar{x}\|^2$$

A matrix with $A^\top A = I$ is **orthogonal**, and it keeps every length.

What it says about the columns

Entry (i, j) of $A^\top A$ is column i dotted with column j. So:

- every column has **length** 1;
- the columns are **perpendicular**.

An **orthonormal basis** (Unit 2-2).

Check: $R(\theta)$

- Lengths: $\cos^2\theta + \sin^2\theta = 1$.
- Dot product: $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$.

Every rotation is orthogonal.

Notebook `U2-4_Transformations-2_NamedTransforms`, §2, lengths measured after each matrix

Sorting the Types

Orthogonal (lengths kept)

- **Rotation** $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$: columns of length 1, perpendicular; the F turns.
- **Reflection** $\left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$: the columns are $\hat{\imath}$ and $-\hat{\jmath}$; the F flips.
- **Permutation** $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$: the columns are $\hat{\jmath}$ and $\hat{\imath}$; the coordinates swap.

Not orthogonal (lengths change)

- **Scaling** $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 1/2 \end{smallmatrix}\right]$: $A^\top A = \left[\begin{smallmatrix} 4 & 0 \\ 0 & 1/4 \end{smallmatrix}\right] \neq I$; lengths along x double.
- **Shear** $\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$: the second column has length $\sqrt{2}$; the top of the F got longer.
- **Projection** $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$: the second column is $\bar{0}$; $\hat{\jmath}$ loses its length.

Orthogonal matrices keep angles too ($\bar{u}^\top A^\top A \bar{v} = \bar{u}^\top \bar{v}$): shapes come out **congruent**.

Why data science cares: PCA's change of basis is orthogonal, so distances between data points are unchanged: a rotation, not a distortion.

Exercise: Orthogonal or Not?

Try it

For each matrix, compute $A^\top A$ and say whether the transformation preserves length. For the ones that do, say what they do geometrically.

$$A_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} \frac{3}{5} & -\frac{4}{5} \\ \frac{4}{5} & \frac{3}{5} \end{bmatrix}.$$

Then: without any computation, is $2A_1$ orthogonal?

Exercise: Solution

Answers

1. $A_1^\top A_1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. **Orthogonal**: it is the quarter-turn rotation $R(90°)$.
2. $A_2^\top A_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \neq I$. **Not orthogonal**: the shear stretches its second column to length $\sqrt{2}$.
3. Column lengths: $\sqrt{\frac{9}{25} + \frac{16}{25}} = 1$ for both; dot product $-\frac{12}{25} + \frac{12}{25} = 0$. So $A_3^\top A_3 = I$: **orthogonal**. It is $R(\theta)$ with $\cos\theta = \frac{3}{5}$, $\sin\theta = \frac{4}{5}$, a rotation by about $53°$. (The 3-4-5 triangle again.)
4. **No.** $2A_1$ sends $\hat{\imath}$ to a vector of length 2. Algebraically $(2A_1)^\top (2A_1) = 4A_1^\top A_1 = 4I \neq I$. Scaling any orthogonal matrix by $k \neq \pm 1$ breaks the property.

Composition and Order

Two Moves in a Row Is One Matrix

Composition is the product: B first, then A

$$\bar{x} \longrightarrow B\bar{x} \longrightarrow A(B\bar{x}) = (AB)\,\bar{x}$$

- Two moves are **one matrix**, AB: the **first** move sits on the **right**.

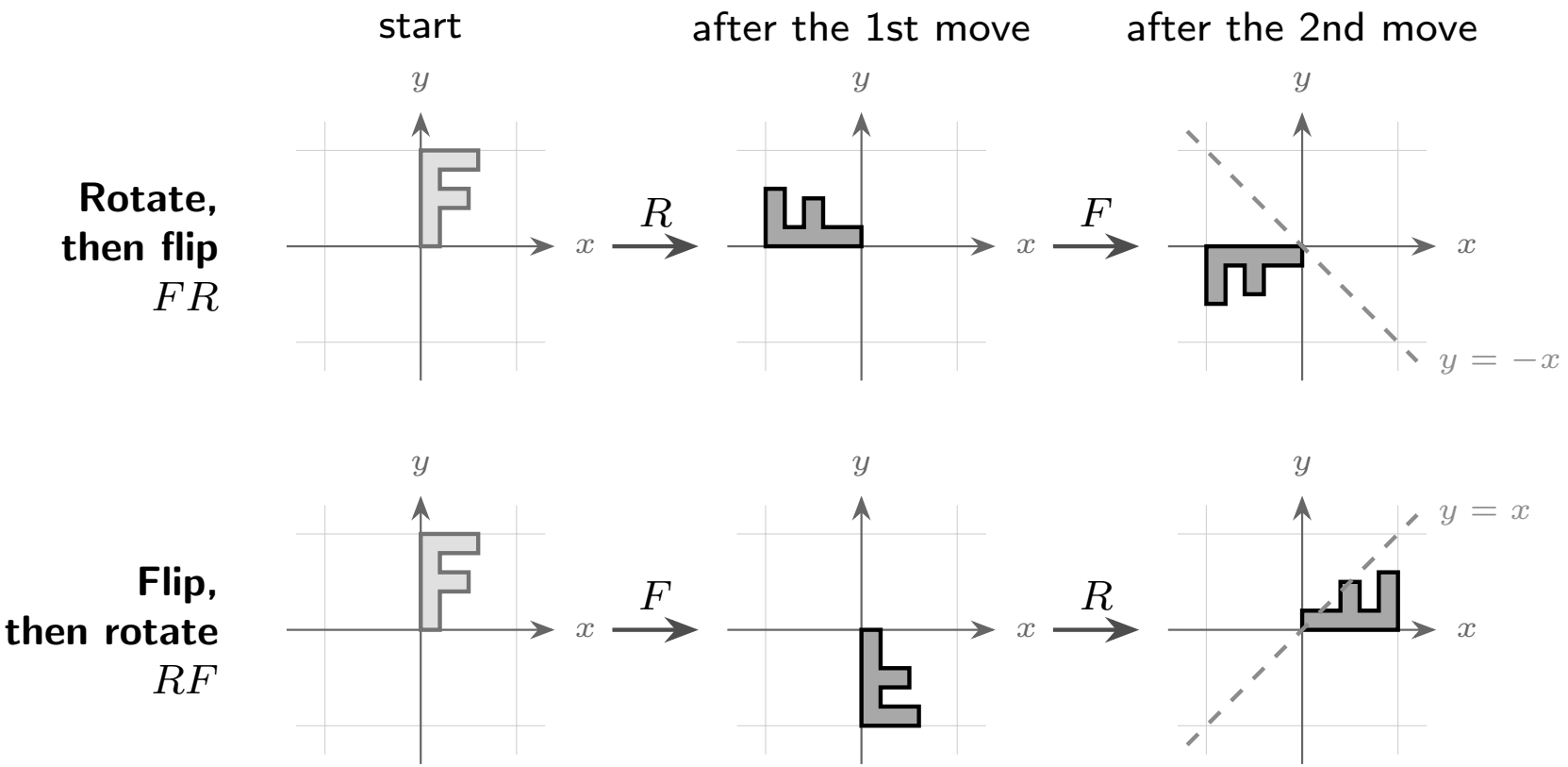
Worked: quarter turn $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, flip across the x-axis $F = \left[\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right]$

rotate, then flip: $FR = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$ flip, then rotate: $RF = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

- FR: a reflection across $y = -x$.
- RF: swaps the coordinates, a reflection across $y = x$.
- **Different.** $AB \neq BA$ from Unit 2-3, now as a picture.

Notebook `U2-4_Transformations-2_NamedTransforms`, §3, both orders side by side

The Two Orders, Drawn



Same two moves, opposite order, different final position: FR reflects across $y = -x$, RF across $y = x$. Read the last F's columns in each row and you recover the product computed on the previous slide.

Powers of Transformations

Applying one transformation n times is the matrix power A^n from Unit 2-3, and knowing the type of transformation makes the powers easy to predict.

Powers of each type

Scaling	$\left[\begin{smallmatrix} a & 0 \\ 0 & b \end{smallmatrix}\right]^n = \left[\begin{smallmatrix} a^n & 0 \\ 0 & b^n \end{smallmatrix}\right]$	**accumulates**
Shear	$\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^n = \left[\begin{smallmatrix} 1 & n \\ 0 & 1 \end{smallmatrix}\right]$	**accumulates**
Rotation	$R(\theta)^n = R(n\theta)$, so $R(90^\circ)^4 = I$	**cycles**
Reflection	$F^2 = I$	**cycles**
Permutation	swap: $P^2 = I$; 3-cycle: $P^3 = I$	**cycles**
Projection	$P^n = P$	**stays put**

Exercise: Composition and Powers

Try it

Let $R = R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

1. Write the single matrix for "double everything, then turn a quarter turn." Then the matrix for the opposite order. Are they different? Explain why in this case.
2. Compute R^2 and say in words what it does. What is R^3 as a rotation?
3. Compute PR and RP and describe each geometrically. Which order projects onto the y-axis?

Exercise: Solution

Answers

1. Double first, then turn: $RS = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. Turn first, then double: $SR = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. **The same**, because $S = 2I$ and a multiple of the identity commutes with everything: $R(2I) = 2R = (2I)R$. Non-commutativity is the general rule, but uniform scaling is the classic exception.
2. $R^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$: two quarter turns is a **half turn**, every point sent to its opposite. R^3 is three quarter turns, a rotation by $270°$, the same as a quarter turn clockwise.
3. $PR = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$: turn first, then flatten onto the x-axis. $RP = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$: flatten onto the x-axis first, then turn that axis up onto the y-**axis**. So RP is the one that lands everything on the y-axis. Check on $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$: RP gives $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$.

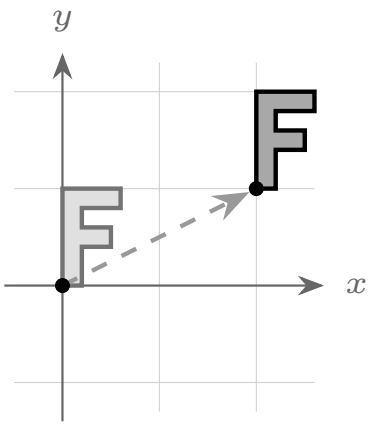
Moving Away from the Origin

Matrices in Graphics and Robotics

Three Moves No Matrix Can Make

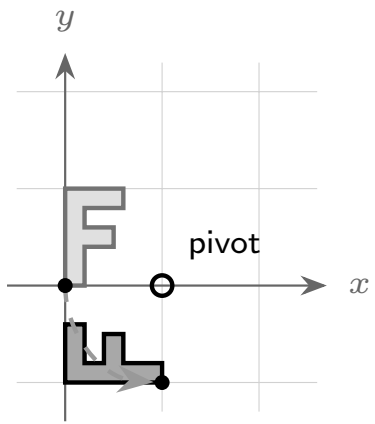
Every move so far turned, flipped, or stretched the plane **about the origin**. Graphics and robots also need these three, and no matrix does any of them:

Slide by $(2, 1)$



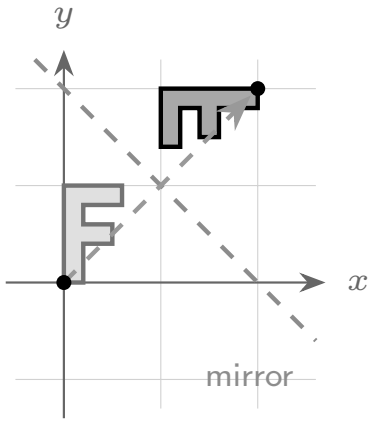
origin $\to (2, 1)$

Turn about $(1, 0)$



origin $\to (1, -1)$

Reflect across $x + y = 2$



origin $\to (2, 2)$

In all three, the **origin moves**. The next slide shows that this is exactly what a matrix can never do.

Why No Matrix Can Do It

Multiply any matrix by the zero vector

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} a \cdot 0 + b \cdot 0 \\ c \cdot 0 + d \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Whatever the entries, every row times a column of zeros is 0. So **every matrix keeps the origin exactly where it is**.

So none of the three moves is a matrix

- **Slide:** every point moves, the origin included.
- **Turn about another point:** the origin swings around the pivot.
- **Reflect across a line that misses the origin:** the origin lands on its mirror image.

Turning about the origin, or reflecting across a line through it, is fine: the origin stays put.

The robot arm needs exactly this: the forearm turns about the **elbow**, not the origin. Next: one extra coordinate gets around the problem.

The Trick: One Extra Coordinate

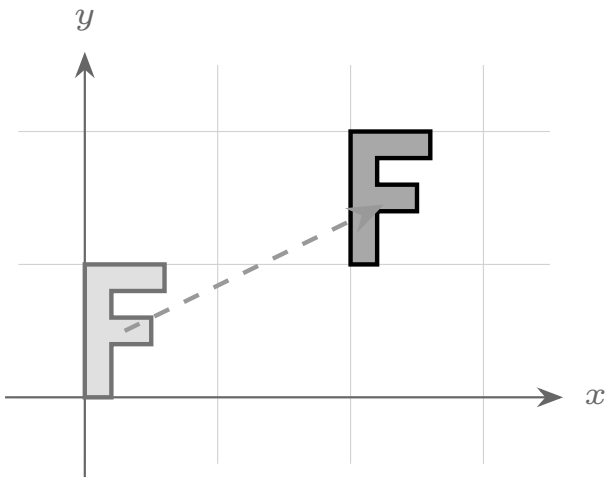
Write (x, y) as $(x, y, 1)$: a slide by $\bar{t} = \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$ becomes a matrix

$$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_1 \\ y + t_2 \\ 1 \end{bmatrix} \qquad \text{e.g.} \quad \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

Sliding by $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ sends the point $(1, 0)$ to $(3, 1)$.

Why this is allowed

In 3D this matrix is a **shear**: perfectly linear, and it fixes $(0, 0, 0)$. We only look at the slice where the last coordinate is 1, and there the shear looks like a slide. These are **homogeneous coordinates**.



Every point of the F slides by $(2, 1)$.

Where This Runs Every Day

The extra coordinate is everywhere

- **Video games and 3D graphics.** Points get a fourth coordinate, $(x, y, z, 1)$, and every move is a 4×4 matrix. A graphics card multiplies millions of points by these matrices every frame.
- **Robot arms.** Each joint is one such matrix, and the gripper's position is the product along the chain of joints (**forward kinematics**).
- **Medical imaging.** A common MRI file format (NIfTI) stores a 4×4 matrix that sends each voxel's index to its position in the scanner. Lining up two scans means finding one such matrix.
- **Image augmentation.** Shifting a picture, not just turning it, uses the same trick.

The idea in one line: **add a dimension, and a move that was not linear becomes linear**.

Every Move Is One 3×3 Matrix

Block form: do A, then slide by $\bar{t}$

$$\begin{bmatrix} A & \bar{t} \\ \bar{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \bar{x} \\ 1 \end{bmatrix} = \begin{bmatrix} A\bar{x} + \bar{t} \\ 1 \end{bmatrix} \qquad \text{e.g. } \begin{bmatrix} a & b & t_1 \\ c & d & t_2 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiply it out row by row to check. A plain matrix has $\bar{t} = \bar{0}$; a plain slide has $A = I$.

Rotate about a point $\bar{p}$, not the origin: three moves, one product

$$\begin{bmatrix} I & \bar{p} \\ \bar{0}^\top & 1 \end{bmatrix} \quad \begin{bmatrix} R & \bar{0} \\ \bar{0}^\top & 1 \end{bmatrix} \quad \begin{bmatrix} I & -\bar{p} \\ \bar{0}^\top & 1 \end{bmatrix}$$

- **Read from right to left:** slide $\bar{p}$ to the origin, rotate, then slide back. A quarter turn about $\bar{p} = (1, 1)$ keeps $(1, 1)$ fixed and sends $(2, 1)$ to $(1, 2)$.
- This is the robot arm's **elbow**: the forearm turns about the elbow, not the base. Next slide.

Back to the Robot Arm

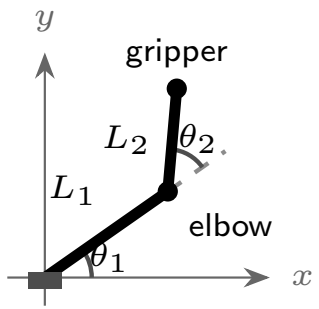
The catch: the forearm turns about the **elbow**, not the origin.

Two 3×3 matrices: R_θ turns about the origin; T_L slides right by L

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad T_L = \begin{bmatrix} 1 & 0 & L \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Elbow at the origin first, then move it out

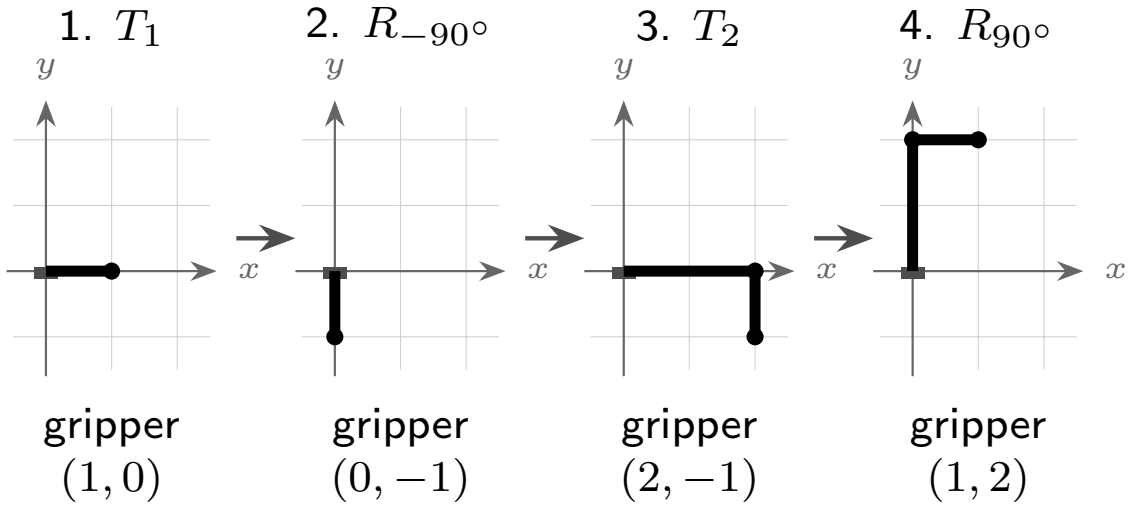
1. T_{L_2}: forearm along the x-axis, elbow at the origin.
2. R_{θ_2}: bend at the elbow, which **is** the origin now.
3. T_{L_1}: push it out to the end of the upper arm.
4. R_{θ_1}: swing the whole arm about the base.



$$\text{gripper} = R_{\theta_1}\, T_{L_1}\, R_{\theta_2}\, T_{L_2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \qquad \text{(step 1 sits on the right)}$$

The Robot Arm, Step by Step

$L_1 = 2$, $L_2 = 1$, $\theta_1 = 90°$, $\theta_2 = -90°$: apply the four matrices to $(0, 0, 1)$, right to left.



All four at once: $R_{90°}\, T_2\, R_{-90°}\, T_1$

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Times $(0, 0, 1)$ this picks out the **last column**, $(1, 2)$: the gripper, as drawn.

The Whole Arm as One Matrix

The same four matrices for **any** angles and lengths, in order R_{θ_1}, T_{L_1}, R_{θ_2}, T_{L_2}:

$$\begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & L_1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 \\ \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & L_2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) & L_1\cos\theta_1 + L_2\cos(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) & L_1\sin\theta_1 + L_2\sin(\theta_1+\theta_2) \\ 0 & 0 & 1 \end{bmatrix}$$

Reading the result

- **Last column: the gripper.** The elbow $(L_1\cos\theta_1, L_1\sin\theta_1)$, plus the forearm's L_2 at the total angle $\theta_1+\theta_2$.
- **Top-left block: the 2×2 rotation by $\theta_1+\theta_2$.** Slides never change direction, so the two turns add.
- **Check:** $L_1=2$, $L_2=1$, $\theta_1=90°$, $\theta_2=-90°$ gives the matrix on the previous slide.

Review: Transformations

What we covered: a matrix as a map

1. Multiplying by A is a **function** on vectors, $f(\bar{x}) = A\bar{x}$ (a **linear transformation**): it moves every point of the plane by the same rule.
2. The rule is **linear**: $A(\bar{u} + \bar{v}) = A\bar{u} + A\bar{v}$ and $A(c\bar{u}) = cA\bar{u}$. So the origin is fixed, lines stay lines, and grids stay evenly spaced. Translation is not linear.
3. One product, **two interpretations**: $B\bar{c}$ is a **new vector** in the same space, or the **same vector** with its $\bar{b}$-basis address $\bar{c}$ rewritten in standard coordinates.
4. **The columns of A are where $\hat{\imath}$ and $\hat{\jmath}$ land.** Read any matrix as a picture; build any transformation by choosing two images and stacking them.
5. The **types of transformations**: scaling (diagonal), shear, rotation $R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ (and R_x, R_y, R_z in 3D), reflection, permutation (a row swap is one), and projection, which destroys information. Onto any line, projection is the **outer product** $P = \hat{w}\,\hat{w}^\top$, and the leftover is $(I - P)\,\bar{v}$.

Review: Transformations

What we covered: length and order

6. A transformation **preserves length** exactly when $A^\top A = I$ (**orthogonal**): its columns are an orthonormal basis. Rotations, reflections, and permutations qualify; scaling, shear, and projection do not.
7. **Two moves in a row are the product** AB, first move on the right. Order matters: rotate-then-flip and flip-then-rotate are different reflections.
8. **Powers** repeat a move: rotations cycle, a projection is idempotent ($P^2 = P$), shears and scalings accumulate, a reflection undoes itself.

Review: Transformations

What we covered: moving away from the origin

9. **A matrix never moves the origin**, since $A\bar{0} = \bar{0}$. So no matrix can slide the plane, or turn or reflect it about anything other than the origin.
10. **One extra coordinate**: write (x, y) as $(x, y, 1)$. A slide becomes a 3×3 matrix, every move is $\begin{bmatrix} A & \bar{t} \\ \bar{0}^\top & 1 \end{bmatrix}$, and turning about any point is one product.
11. **The robot arm**: build it with the elbow at the origin, then move it out. Gripper $= R_{\theta_1} T_{L_1} R_{\theta_2} T_{L_2}$ applied to $(0, 0, 1)$. 3D graphics does the same with 4×4 matrices.

Notebooks: `U2-4_Transformations-1`, `-2`, `-3`. **Previous:** Unit 2-3, Matrices.
Next: Unit 2-5, **Determinants and Inverses**.